

# INITIAL ANALYSIS DOCUMENT

## HealthConnect Experience Lab — Data Analytics Track | Week 4

*Analytical Foundation for the Investigation of Missed Appointments*

### WEEK 4 OBJECTIVE

Understand the HealthConnect appointment data, assess its quality, define the Business Questions and hypotheses to investigate, identify potential KPIs, and prepare a structured analytical approach — without prematurely executing the full EDA or drawing causal conclusions.

## 1. PROJECT CONTEXT & BUSINESS PROBLEM

HealthConnect Clinic faces missed appointments, cancellations, and patient support needs. Within the Experience Lab, the broader question is how data and AI can help reduce missed appointments and improve the patient support experience. For the Data Analytics Track, Week 4 focuses on understanding the data and preparing the analytical investigation of no-shows.

**Analytical problem: understand the extent of no-shows and identify, within the available variables, the contexts and patterns that warrant deeper investigation to inform future operational decisions.**

**Interpretation principle: the available data are observational. Future relationships will therefore be described as associations or observed differences unless a subsequent analytical design supports a stronger conclusion.**

## 2. DATA DICTIONARY REVIEW

The Data Dictionary was used as the definition reference before any analysis. It confirms 18 variables covering appointment identification, recorded patient characteristics, scheduling context, recorded history, reminders, accessibility, and the final appointment outcome.

| Architecture               | Variables                                                                                              | Initial Analytical Role                            |
|----------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Identifiers                | appointment_id, patient_id                                                                             | Appointment grain and anonymized patient reference |
| Recorded Characteristics   | gender, age, age_group                                                                                 | Descriptive segmentation                           |
| Appointment & Scheduling   | appointment_type, booking_date, appointment_date, appointment_day, appointment_time, booking_lead_days | Appointment context and booking lead time          |
| Recorded History           | previous_appointments, previous_no_shows                                                               | Recorded summary historical information            |
| Reminder                   | reminder_sent, reminder_channel                                                                        | Reminder coverage and channel                      |
| Accessibility / Operations | distance_to_clinic_km, waiting_time_minutes                                                            | Approximate distance and estimated waiting time    |
| Outcome                    | appointment_outcome                                                                                    | Final status: Attended, No-Show, Cancelled         |

## 3. DATASET OVERVIEW

| Item                    | Result                         | Interpretation                          |
|-------------------------|--------------------------------|-----------------------------------------|
| Number of rows          | 5,000                          | Each row represents one appointment     |
| Number of columns       | 18                             | Consistent with the Data Dictionary     |
| Analytical grain        | Appointment record             | Selected unit of analysis               |
| appointment_id          | 5,000 unique values            | No duplicate appointment identifiers    |
| patient_id              | 1,696 unique values            | May appear across multiple appointments |
| appointment_date period | 01 Jan 2025 – 30 Jun 2026      | Scheduled appointment window            |
| Outcome                 | Attended / No-Show / Cancelled | Primary outcome variable                |

The selected analytical grain is therefore the appointment rather than the patient. This distinction is important for analyses involving repeated patient\_id values because some attributes associated with these identifiers are not longitudinally stable in the dataset.

## 4. INITIAL DATA QUALITY ASSESSMENT

The purpose of this stage is to determine whether the dataset can support the planned analysis and to identify the precautions required before any KPI calculation or hypothesis testing.

| Check                 | Result                                                                                              | Assessment     | Analytical Handling                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------|
| Structure             | 5,000 × 18; all 18 expected variables present                                                       | PASS           | Retain schema                                                             |
| appointment_id        | 5,000 unique; 0 duplicates                                                                          | PASS           | Appointment-level grain confirmed                                         |
| Dates / Lead Time     | 0 booking_date > appointment_date; 0 booking_lead_days mismatches; 0 appointment_day mismatches     | PASS           | Convert date fields to Date type before analysis                          |
| Reminder channel      | 1,366 missing values; all correspond to reminder_sent = No; 0 reminder_sent = Yes without a channel | STRUCTURAL N/A | Treat as Not Applicable, not as a data-quality defect                     |
| distance_to_clinic_km | 90 missing values (1.8%)                                                                            | ATTENTION      | Assess handling before BQ4; no automatic imputation                       |
| waiting_time_minutes  | 60 missing values (1.2%)                                                                            | ATTENTION      | Assess handling before BQ4; no automatic imputation                       |
| Internal History      | 0 rows with previous_no_shows > previous_appointments                                               | PASS           | Logical relationship respected at row level                               |
| Repeated patient_id   | Inconsistencies observed in age, gender, and historical variables                                   | LIMITATION     | No reliable longitudinal reconstruction; appointment-level interpretation |

## 5. RELEVANT VARIABLES IDENTIFICATION

The final selection is based not on column availability but on each variable's contribution to a Business Question. Identifiers are retained for structure and validation, while analytical variables are organized according to the investigation architecture.

| Analytical Architecture    | Variables                                                  | Role                                                 | Link    |
|----------------------------|------------------------------------------------------------|------------------------------------------------------|---------|
| Outcome / Baseline         | appointment_outcome                                        | Core outcome variable                                | BQ1     |
| Appointment Context        | appointment_type                                           | Compare patterns by appointment type                 | BQ2     |
| Scheduling Context         | appointment_day, appointment_time, booking_lead_days       | Examine scheduling-related differences               | BQ2     |
| Reminder & Engagement      | reminder_sent, reminder_channel                            | Examine patterns associated with reminders           | BQ3     |
| Accessibility & Operations | distance_to_clinic_km, waiting_time_minutes                | Examine accessibility and operational context        | BQ4     |
| Recorded History           | previous_appointments, previous_no_shows                   | Explore recorded history cautiously                  | BQ5     |
| Recorded Characteristics   | age, age_group, gender                                     | Descriptive segmentation at appointment-record level | BQ6     |
| Structural Support         | appointment_id, patient_id, booking_date, appointment_date | Validation, grain, derivations, and traceability     | Support |

## 6. RELEVANT BUSINESS QUESTIONS

| Business Question                      | Question to Investigate                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| BQ1 — Overall Appointment Outcome      | What is the overall distribution of appointment outcomes, and what is the relative extent of no-shows?                                                    |
| BQ2 — Appointment & Scheduling Context | How do no-show patterns vary across appointment types and scheduling characteristics, including appointment day, appointment time, and booking lead time? |
| BQ3 — Reminder & Engagement            | How do no-show patterns vary by reminder status and, where applicable, by                                                                                 |

| Business Question                         | Question to Investigate                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                                           | the reminder channel used?                                                                                    |
| BQ4 — Accessibility & Operational Context | How do no-show patterns vary across recorded distance-to-clinic and estimated waiting-time levels?            |
| BQ5 — Recorded Patient History            | How do no-show patterns vary according to the recorded number of previous appointments and previous no-shows? |
| BQ6 — Recorded Patient Characteristics    | How do no-show patterns vary across recorded age groups and gender categories?                                |

## 7. ANALYTICAL HYPOTHESES — VALUE-ADDED

The hypotheses below structure future comparisons. They are formulated as relationships to be tested, not as established findings. BQ1 provides the descriptive baseline and does not require a formal hypothesis.

| Hypothesis | Variable / Focus      | Relationship to Test                                                                   |
|------------|-----------------------|----------------------------------------------------------------------------------------|
| H2a        | Appointment Type      | The no-show rate differs across appointment types.                                     |
| H2b        | Appointment Day       | The no-show rate differs across appointment days.                                      |
| H2c        | Appointment Time      | The no-show rate differs across appointment time periods.                              |
| H2d        | Booking Lead Time     | No-show patterns differ according to the time between booking and appointment.         |
| H3a        | Reminder Sent         | The no-show rate differs between appointments with and without a recorded reminder.    |
| H3b        | Reminder Channel      | Among appointments with a reminder, the no-show rate differs across reminder channels. |
| H4a        | Distance to Clinic    | No-show patterns differ according to the recorded distance to the clinic.              |
| H4b        | Waiting Time          | No-show patterns differ according to estimated waiting time.                           |
| H5a        | Previous Appointments | No-show patterns differ according to the recorded number of previous appointments.     |
| H5b        | Previous No-Shows     | No-show patterns differ according to the recorded number of previous no-shows.         |
| H6a        | Age Group             | The no-show rate differs across recorded age groups.                                   |
| H6b        | Gender                | The no-show rate differs across recorded gender categories.                            |

## 8. 3 POTENTIAL KPIS

Following the Business Question → Decision Need → Potential KPI logic, only three indicators are retained at this stage. Segmentation measures required for BQ2–BQ6 will remain analytical metrics unless a clear decision or monitoring need justifies their promotion to KPI status.

| Potential KPI                   | BQ  | Decision Use                                                                       | Week 4 Status                                                  |
|---------------------------------|-----|------------------------------------------------------------------------------------|----------------------------------------------------------------|
| No-Show Rate (%)                | BQ1 | Measure and, subsequently, monitor the relative magnitude of missed appointments   | Potential KPI — denominator to be finalized before calculation |
| Reminder Coverage Rate (%)      | BQ3 | Measure the operational coverage of the reminder process                           | Potential KPI                                                  |
| No-Show Rate by Reminder Status | BQ3 | Compare no-show patterns between appointments with and without a recorded reminder | Potential diagnostic / decision-support KPI — association only |

Methodological note: the No-Show Rate will only be calculated after explicitly defining the eligible population, numerator, and denominator. In particular, Cancelled appointments must be handled according to a documented analytical convention before any reporting.

## 9. INITIAL ANALYSIS APPROACH

### 9.1 Analytical Objective

The analysis will investigate patterns associated with missed appointments using the available appointment-level data. It will first establish the overall extent of no-shows, then examine variations across appointment and scheduling context, reminder practices, accessibility and operational context, recorded patient history, and recorded patient characteristics.

### 9.2 Analytical Scope

Unit of analysis: appointment record. Primary outcome: appointment\_outcome, with a primary focus on No-Show. Longitudinal patient-level interpretation will be limited because inconsistencies were identified in some attributes associated with repeated patient\_id values.

### 9.3 Business Question → Hypothesis → Analysis Map

| BQ  | Focus                      | Hypotheses  | Planned Analysis                                                |
|-----|----------------------------|-------------|-----------------------------------------------------------------|
| BQ1 | Baseline                   | Descriptive | Outcome distribution; No-Show Rate after denominator definition |
| BQ2 | Appointment & Scheduling   | H2a–H2d     | Comparisons by type, day, time period, and lead time            |
| BQ3 | Reminder & Engagement      | H3a–H3b     | Reminder Yes vs No; then conditional comparison across channels |
| BQ4 | Accessibility & Operations | H4a–H4b     | Patterns across distance and waiting time                       |
| BQ5 | Recorded History           | H5a–H5b     | Comparisons across recorded history                             |
| BQ6 | Recorded Characteristics   | H6a–H6b     | Comparisons across age group and gender                         |

### 9.4 Proposed Analytical Methods

| Analytical Need                | Proposed Method                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Overall profile                | Frequencies and proportions                                                                 |
| Categorical segment comparison | Counts, rates, cross-tabulations, and group-level comparisons                               |
| Continuous / ordinal variables | Distribution analysis followed by grouping only where justified                             |
| Reminder status                | Comparison of No-Show Rates                                                                 |
| Reminder channel               | Conditional comparison where reminder_sent = Yes                                            |
| Magnitude of differences       | Absolute and relative differences where appropriate                                         |
| Statistical evidence           | Appropriate tests only where justified by the question, variable type, and test assumptions |

### 9.5 Potential KPI Framework

The three potential KPIs in Section 8 form the initial measurement framework. Additional segmentation metrics will support BQ2–BQ6 but will not automatically be classified as KPIs.

### 9.6 Data Quality Considerations

Future analyses will explicitly account for missing values in distance and waiting time, the Not Applicable nature of reminder\_channel when reminder\_sent = No, and the longitudinal limitations of patient\_id. No automatic imputation or arbitrary categorization will be applied before reviewing the distribution and analytical need.

## 9.7 Interpretation Principles

### Finding → Interpretation → Business Implication → Recommendation

Future recommendations must be linked to identifiable analytical evidence. An observed association may justify further investigation, validation, or a pilot; it will not automatically be presented as a causal relationship.

## 10. ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

| Type       | Key Consideration                                                                            | Analytical Response                                                             |
|------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Assumption | Each row represents one appointment and appointment_outcome its recorded final status.       | Maintain appointment-level analytical grain.                                    |
| Assumption | Historical variables may be used as recorded information at row level.                       | Do not reconstruct longitudinal patient histories.                              |
| Limitation | Demographic and historical inconsistencies exist across some repeated patient_id values.     | Limit patient-level conclusions.                                                |
| Limitation | Distance and waiting time contain limited missingness.                                       | Document handling before BQ4 analysis.                                          |
| Limitation | The dataset is observational and limited to the variables provided.                          | Do not infer causality or rule out unobserved determinants.                     |
| Risk       | An inappropriate denominator could change the interpretation of the No-Show Rate.            | Define population, numerator, denominator, and formula before calculation.      |
| Risk       | Arbitrary grouping of lead time, distance, or waiting time could create misleading patterns. | Review distributions and justify any analytical bands.                          |
| Risk       | Segmentation metrics could be incorrectly promoted to KPIs.                                  | Require an identifiable decision or monitoring need.                            |
| Dependency | Final KPI and recommendation selection will depend on future findings.                       | Do not pre-select priority segments.                                            |
| Dependency | Future consistency depends on stable metric definitions.                                     | Build a Metric Reference Table / Single Source of Truth before final reporting. |

## 11. WEEK 4 CONCLUSION & NEXT STEPS

Week 4 establishes a structured analytical foundation for the HealthConnect Experience Lab. The dataset and Data Dictionary were reviewed, the analytical grain and key data-quality constraints were identified, relevant variables were organized, six Business Questions and twelve analytical hypotheses were defined, and three potential KPIs were selected using a decision-oriented logic.

No analytical finding regarding no-show drivers is claimed at this stage. Subject to the official Week 5 instructions, the logical next step will be to prepare the data, finalize metric definitions and analytical populations, execute the planned analyses, test the hypotheses, and translate only validated findings into business implications and recommendations.

**WEEK 4 FOUNDATION COMPLETE**  
UNDERSTAND → REVIEW → DEFINE → PLAN

**Nancy Lee YIMBERE ALAPINI**

*Performance & Decision Intelligence Analyst*